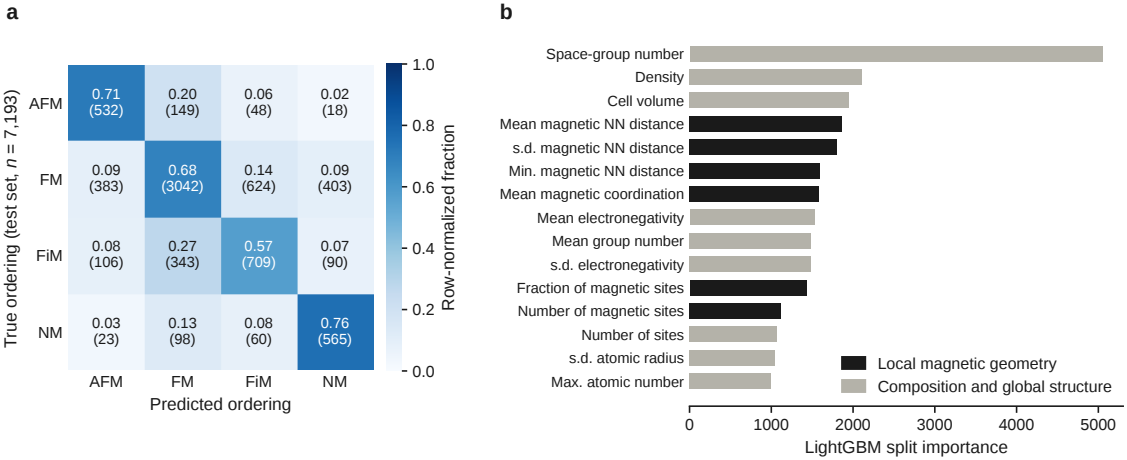
a
True ordering (test set, n = 7,193)
AFM
FM
FiM
NM
0.71 (532)
0.20 (149)
0.06 (48)
0.02 (18)
0.09 (383)
0.68 (3042)
0.14 (624)
0.09 (403)
0.08 (106)
0.27 (343)
0.57 (709)
0.07 (90)
0.03 (23)
0.13 (98)
0.08 (60)
0.76 (565)
AFM
FM
FiM
NM
Predicted ordering
Row-normalized fraction
1.0
0.8
0.6
0.4
0.2
0.0
b
Space-group number
Density
Cell volume
Mean magnetic NN distance
s.d. magnetic NN distance
Min. magnetic NN distance
Mean magnetic coordination
Mean electronegativity
Mean group number
s.d. electronegativity
Fraction of magnetic sites
Number of magnetic sites
Number of sites
s.d. atomic radius
Max. atomic number
Local magnetic geometry
Composition and global structure
0
1000
2000
3000
4000
5000
LightGBM split importance